

Credit Card Financial Analysis Dashboard

Project Report

Prepared using SQL Database Management and Microsoft Power BI

Introduction

This project presents a Credit Card Financial Analysis Dashboard developed to analyze credit card transactions and customer behavior. The project uses SQL for database creation and data loading, and Microsoft Power BI for interactive visualization and business intelligence reporting.

The main objective is to transform raw credit card transaction and customer information into meaningful insights that can support business decision-making.

Project Objectives

The objectives of this project are:

- Design and create a structured database for credit card data.
- Import and organize transaction and customer datasets using SQL.
- Analyze revenue, transactions, customer demographics, and card usage patterns.
- Develop interactive dashboards in Power BI.
- Identify important business trends such as customer segments, revenue contributors, and delinquent account patterns.

Tools and Technologies Used

SQL Database:

Used for creating tables, defining data structures, and importing CSV datasets.

Microsoft Power BI:

Used for data modeling, visualization, KPI creation, and dashboard development.

Datasets:

Credit card transaction data and customer demographic data were used for analysis.

Database Design

A database named 'ccdb' was created. Two major tables were developed.

Credit Card Transaction Table (cc_detail):

This table stores transaction-related information including card category, annual fees, credit limit, transaction amount, transaction count, utilization ratio, expenditure type, interest earned, and delinquent account status.

Customer Detail Table (cust_detail):

This table stores customer information including age, gender, education level, occupation, income, marital status, location, and customer satisfaction score.

Data Processing Workflow

The project workflow consists of the following steps:

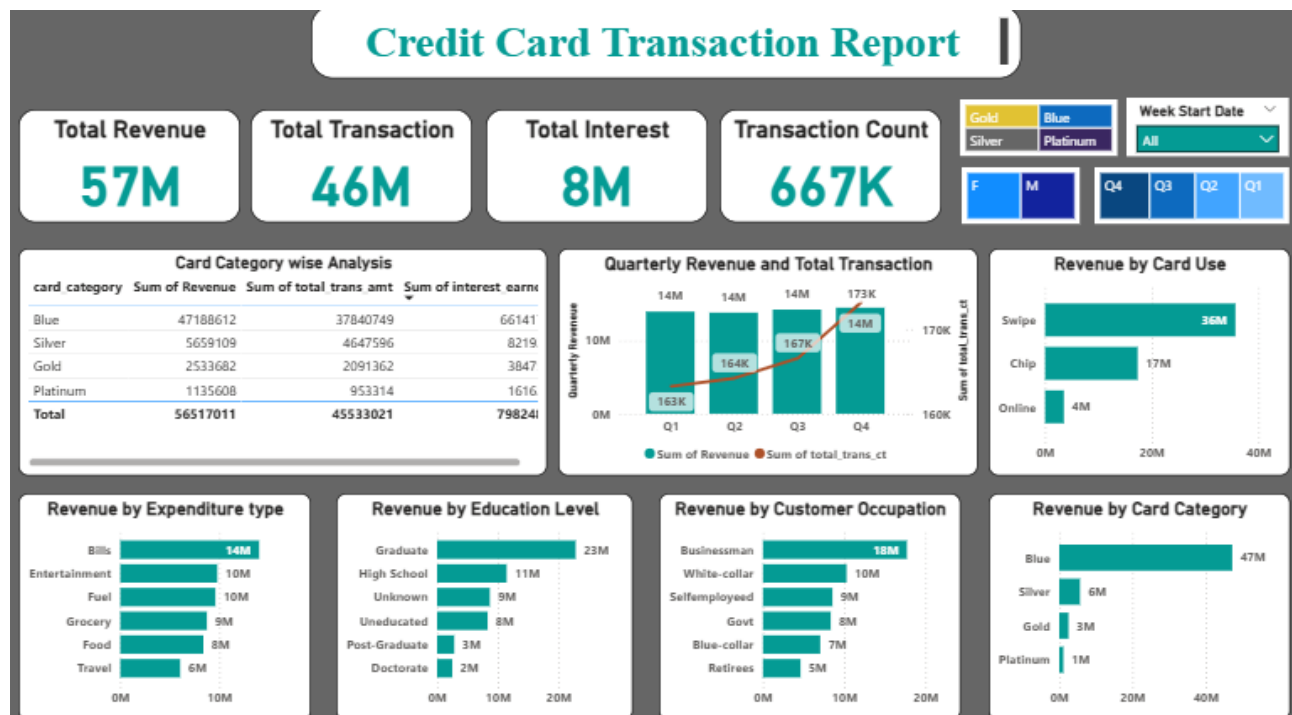
1. Creation of SQL database.
2. Creation of transaction and customer tables.
3. Importing CSV datasets into SQL tables.
4. Connecting the processed data with Power BI.
5. Building analytical dashboards.
6. Extracting business insights from visual reports.

Power BI Dashboard Analysis

The Power BI report contains three major analytical sections:

- A. Credit Card Transaction Report
- B. Credit Card Customer Report
- C. Additional Performance and Risk Analysis

Credit Card Transaction Analysis



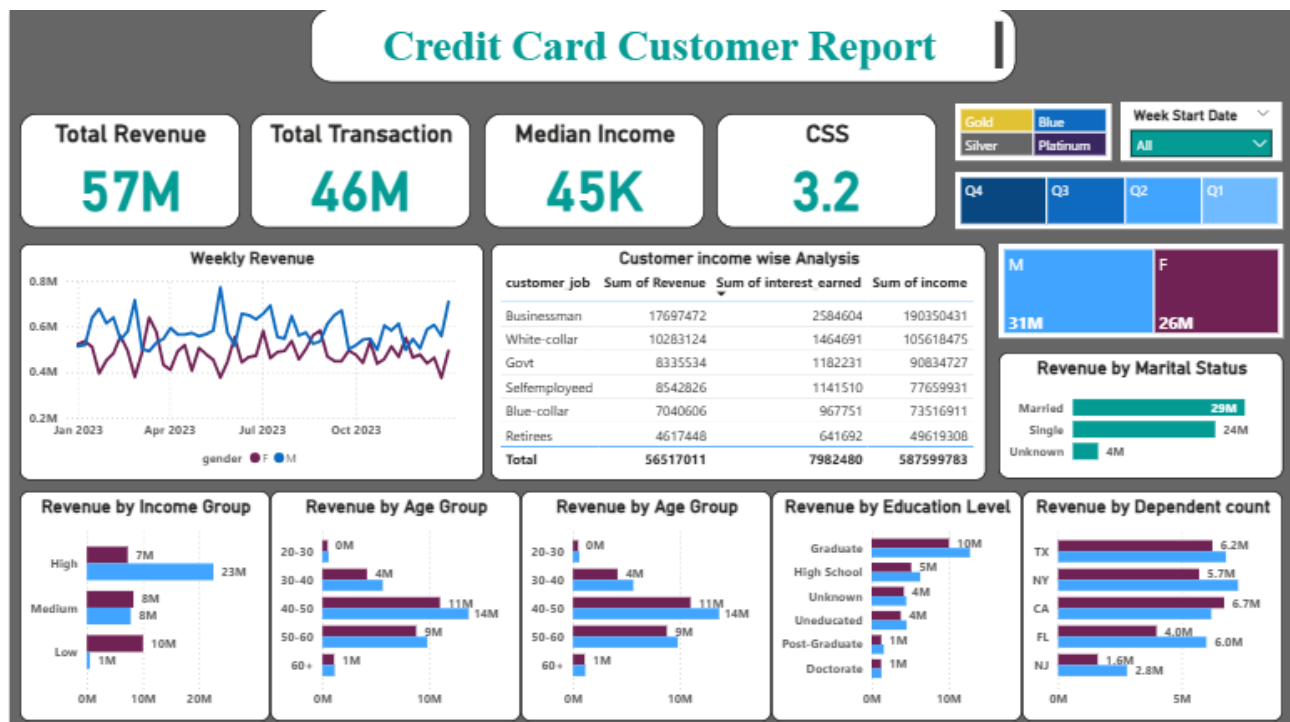
The transaction dashboard provides the following key performance indicators:

- Total Revenue: 57M
- Total Transaction Amount: 46M
- Total Interest Earned: 8M
- Transaction Count: 667K

Major analysis includes revenue by:

- Card category
- Card usage method
- Expenditure type
- Customer occupation
- Education level
- Quarterly performance

Customer Analysis



The customer dashboard evaluates customer characteristics and their relationship with revenue.

Key indicators:

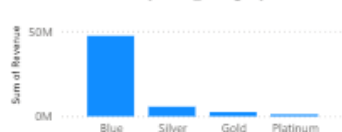
- Total Revenue: 57M
- Total Transaction: 46M
- Median Income: 45K
- Customer Satisfaction Score: 3.2

The report analyzes:

- Income groups
- Age groups
- Gender distribution
- Education levels
- Occupations
- Marital status
- Geographic distribution

Risk and Performance Analysis

Sum of Revenue by card_category



week_num	2	Previous_week Revenue	Current_week Revenue	wow_revenue
53		933134	1201601	28.8%
52		1070439	933134	-12.8%
51		1026549	1070439	4.3%
50		980152	1026549	4.7%
49		1008777	980152	-2.8%
48		1047120	1008777	-3.7%
47		1078915	1047120	-2.9%

Delinquent Account %

delinquent_acc	%GT Count of delinquent
0	93.9
1	6.0
Total	100.0

Transaction count over time



Delinquent by job category

delinquent_acc	Blue-collar	Businessman	Govt	Retirees	Selfemployed	White-collar	Total
0	14.75%	17.81%	14.00%	9.16%	23.87%	14.35%	93.94%
1	0.85%	0.99%	1.10%	0.61%	1.66%	0.85%	6.06%
Total	15.60%	18.80%	15.10%	9.77%	25.53%	15.19%	100.00%

% of Activation Card

activation 30 days	%GT Count of activation 30 days
0	42.54%
1	57.46%
Total	100.00%

Additional analysis includes:

Delinquent Account Analysis:

The dashboard evaluates the percentage of delinquent and non-delinquent accounts.

Card Activation Analysis:

The report analyzes the percentage of cards activated within 30 days.

Weekly Revenue Analysis:

Week-over-week revenue changes are displayed to understand performance trends.

Key Findings

Important observations from the dashboard include:

- Blue card category contributes the largest share of revenue.
- Businessman customers generate the highest revenue contribution among occupations.
- Swipe transactions represent the largest share of card usage.
- Revenue distribution varies across customer income and education groups.
- Delinquency patterns can be monitored based on customer segments.

Future Improvements

Future enhancements may include:

- Customer churn prediction using machine learning.
- Credit risk scoring models.
- Automated data refresh pipelines.
- Advanced customer segmentation.
- Integration with Python-based analytics.

Conclusion

This project demonstrates how SQL and Power BI can be combined to build an effective financial analytics solution. The dashboard converts raw credit card data into interactive visual insights, allowing organizations to understand customer behavior, monitor revenue performance, and evaluate risk indicators.